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## EDUCATION

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**Smith College, Northampton, MA**

*Fall 2022 - Spring 2026*

Double Major in Astronomy & Physics with Honors

### Relevant Coursework

**Astronomy:** Telescopes and Techniques; Stellar Astrophysics; Astronomical Data Science; Cosmology; Independent Study; Observational Techniques in Optical and Infrared Astronomy; Astrophotography & Dark Skies;

**Physics:** Electricity and Magnetism; Newtonian Mechanics; Computational Physics; Mathematical Methods; Light, Relativity, and Quantum Physics; High Precision Spectroscopy; Thermal Physics; Quantum Mechanics; Electronics; Classical Mechanics; Experimental Physics;

**School for Field Studies, South Caicos, Turks and Caicos Islands**

*Fall 2024*

Study Abroad Student at The Center for Marine Resource Studies

**Coursework:** Directed Research; Tropical Marine Ecology; Marine Resource Management; Marine Conservation Governance

## RESEARCH EXPERIENCE

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**Summer Research Internship and Honors Thesis,**  
Five College Astronomy Department

*Summer 2025 - Spring 2026*

**Advisors: Professor James Lowenthal and Professor Min Yun**

- Analyzing JWST NIRCам imaging to quantify the presence of highly obscured active galactic nuclei (AGN) among high redshift galaxies identified as hosts of an AGN using different multi-wavelength tracers. A specific goal of our work was to analyze the morphology of a population of faint radio sources in the COSMOS field in a JWST/NIRCам band and establish the frequency of highly obscured AGN among them. We fit all galaxies with a standard Sérsic profile with a point-source component as well as a double Sérsic profile. A comparison of these models allows us to identify the galaxies in the sample with true point sources.
- Presented [iposter](#) at the 247th meeting of the American Astronomical Society.

**Observational Techniques in Optical and Infrared Astronomy, Smith College**

*Fall 2025*

**Advisor: Professor James Lowenthal**

- We wrote observing proposals to capture images in optical and infrared wavelengths on our 20" telescope for about ninety minutes. I wrote a pipeline for my group's target to reduce the data manually. Writing the pipeline included creating a master bias, master dark, and master flat and subtracting the calibration images from the stacks of the observational images. We also produced a three-color RGB image of our target, M29. Using a standard star we made absolute magnitude measurements of the stars in our cluster, created a CMD of the cluster, and plotted isochrones to estimate the age, distance,

and metallicity of the target. Then we checked published data on M29 to confirm our measurements and identify any errors in our process.

**Independent Study**, Five College Astronomy Department

*Spring 2025*

**Advisors: Professor Min Yun and Professor James Lowenthal**

- Searched for active galactic nuclei as point sources in NIRCам F444W filter of the ultra deep continuum data from COSMOS-Web Field using CARTA for visual inspection and comparison to data from the Very Large Array.

**Directed Research**, School for Field Studies

*Fall 2024*

**Advisor: Professor C.E. O'Brien**

- Developed individual research questions and collected data on the correlation between *Octopus insularis* population and predator abundance via underwater surveys, photo and video analysis. Organized and analyzed data using Microsoft Excel and RStudio. [Presented](#) findings to students, professors, gave a [lightning presentation](#) to the local community, and wrote a [scientific paper](#).

**Investigating Cataclysmic Binaries**, Smith College

*Spring 2024*

- Developed individual research questions and gathered data from VizieR on the influence of mass on stellar parameters of cataclysmic binaries. Conducted data organization and analysis in Python. Presented findings to students, professors, and wrote a [final paper](#).

**High Precision Spectroscopy**, Smith College

*Spring 2023*

**Advisor: Professor Will Raven (Publishing name Will Williams)**

- Gathered absolute frequency measurements on the three excited-state transitions in neutral N14 using saturated absorption spectroscopy.
- Developed an optics setup to improve the signal to noise ratio and designed Arduino based circuit to automate adjusting laser power. Analyzed data in Mathematica, with published results.

**BridgeUP: STEM Brown Scholars Program**,  
American Museum of Natural History, New York City

*Fall 2019 - Summer 2022*

**Advisor: AMNH Research Fellow Danielle Rowland**

- Analyzed data from Transiting Exoplanet Survey Satellite sectors 14 & 15 using Python. Sorted through 32,000 light curves using multiple cuts of the data to find two final exoplanet candidates. Submitted candidates to ExoFOP. Created a [library](#) with functions to sort TESS data using analysis of transit shape.

## **PUBLICATIONS**

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Ahrendsen, K. J., Maruko, C., Albert-Aranovich, K. R., **Berfield-Brewer, Q.**, Esseln, A., Guo, L., Ishimwe, A. E., Kuzniar, Y., McKenna, A. E., Villarreal, K. J., Uprety, A., da Silva, L. V., Vogt, K. F., & Williams, W. D. (2023). Absolute frequency measurement of the  $2p^2(^3P)3s^2P-2p^2(^3P)3p^2D^0$  transitions in neutral  $^{14}\text{N}$ . *Physical Review A*, 108(4).  
<https://doi.org/10.1103/physreva.108.042815>

## ADDITIONAL ACTIVITIES

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### Telescope Maintenance

*Spring 2025 - Spring 2026*

***Employer: Senior Laboratory Instructor Meg Thacher***

- Perform weekly and routine maintenance checks of seven eight-inch Meade telescopes including: RA & DEC motion and locks; finder alignment; telrad alignment and batteries; visual back; diagonal; focus; power supply.

### Astronomy Dept. Open Houses, Smith College

*Spring 2025*

- Operate 8" and 20" telescopes to allow students and community members to view the night sky.

### Hell's Belles Fencing Team, Smith College

*Fall 2022 - Spring 2026*

- Foil Squad Captain: lead and instruct squad at tournaments and practices.
- Foil Fencer: compete in regional and national fencing tournaments and support the team by fundraising.
- Foil Squad Armorer: perform repairs on weapon and body cord circuitry.

## SKILLS

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### Programming Languages

- Proficient in Python, R, Java, Arduino IDE, Mathematica, LaTeX, and HTML

### Software

- ImageJ, LoggerPro, StarStaX, CARTA, SAOImageDS9
- GitHub, Jupyter Notebook, JupyterLab, Google Colab, Visual Studio Code, RStudio
- Microsoft Excel, Microsoft Word, Google Sheets

### Telescope

- Operate telescope for data collection
- Operate telescopes for public stargazing/star parties
- Perform weekly telescope maintenance

### Technical Skills

- Astrophotography with DSLR camera
- General electronics maintenance and circuitry using a breadboard as well as Arduino circuits with sensors and motors. Projects include: AC/DC converter, maze-solving Arduino robot, and an Arduino Simon-says game.
- Video and photo analysis, species identification, visual underwater census (scuba and snorkel), underwater transect surveys
- PADI Advanced Open Water Scuba Certification

## PROFESSIONAL MEMBERSHIPS

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American Astronomical Society

International Dark-Sky Association

Reef Environmental Education Foundation